DO NOW — RECALL

Lesson 085 · Searching, Sorting and Pathfinding Compar

Name: _____ Class: _____ Date: _____

1. What is the sorted-input rule for binary search?

2. State merge sort's worst-time complexity.

3. State f score in A*.

4. Push A,B,C onto a stack, Pop twice, then Push D. Give bottom-to-top order.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The input must be sorted.

2. Model answer

O(n log n).

3. Model answer

f=g+h.

4. Model answer

Pop removes C then B; the final stack is [A,D] bottom-to-top.

By the end, you can…

• Start from the problem and input: searching, ordering and weighted paths are different problem classes.

• A fastest method under one set of assumptions may be invalid or less suitable on another dataset; include preprocessing and memory costs.

• Choose using correctness, worst-case growth, memory, stability where relevant, and an independent trace on new values.

Vocabulary — know these terms

suitability — How well method fits input/task.

precondition — Requirement for correct use.

complexity — Growth of time/space.

heuristic — Estimate guiding search.

NEW LEARNING

Read, annotate and explain

Observe

Start from the problem and input: searching, ordering and weighted paths are different problem classes.

Apply

A fastest method under one set of assumptions may be invalid or less suitable on another dataset; include preprocessing and memory costs.

Evaluate

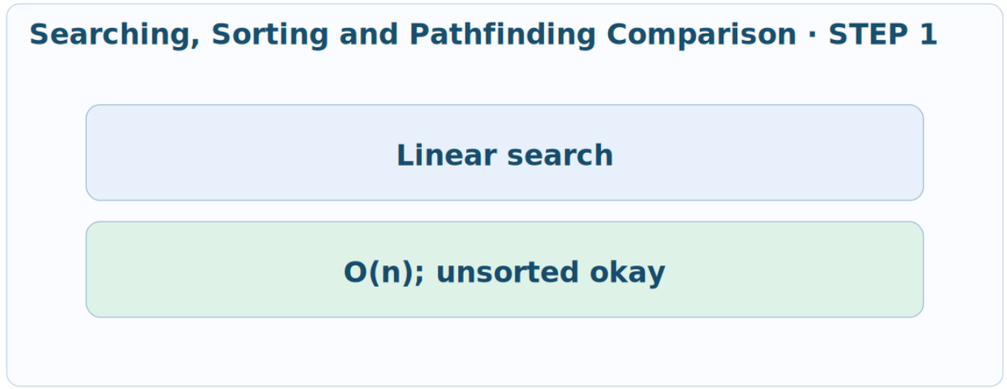
Choose using correctness, worst-case growth, memory, stability where relevant, and an independent trace on new values.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Searching, Sorting and Pathfinding Comparison · STEP 1

New Learning



Searching, Sorting and Pathfinding Comparison · STEP 2

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose method for shortest road journey across a network with non-negative costs but no heuristic.

Dijkstra finds least-cost path with non-negative edge weights; do not confuse with array search.



WEDO · STATE

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

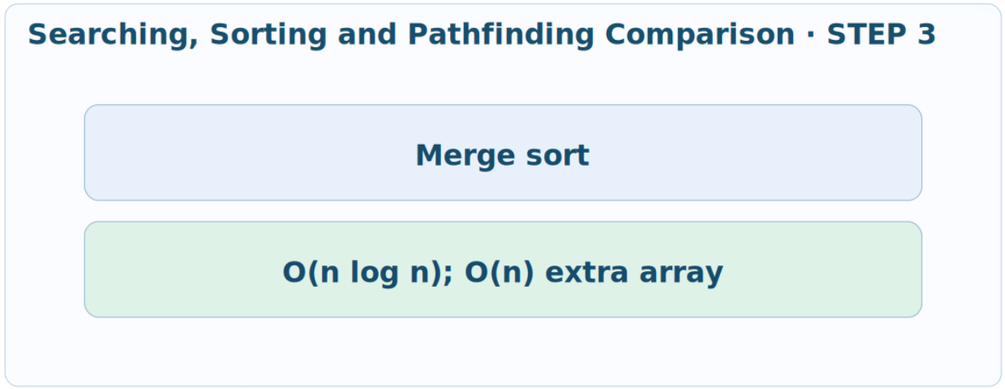
Guided 1 — CHECK

Method and result
Dijkstra finds least-cost path with non-negative edge weights; do not confuse with array search.

New Learning · original lesson diagrams

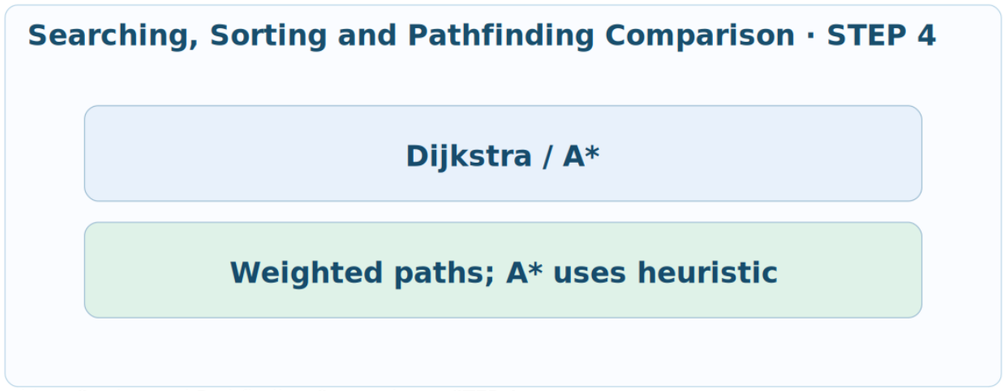
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Searching, Sorting and Pathfinding Comparison · STEP 3

New Learning



Searching, Sorting and Pathfinding Comparison · STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

Choose algorithm for 200 sorted pupils by ID with repeated lookups.

Method and result

Binary search O(log n) is valid because identifiers are sorted; if indexed dictionary is available, keyed lookup is another option.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

All standard algorithms can be swapped freely without checking their input assumptions.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

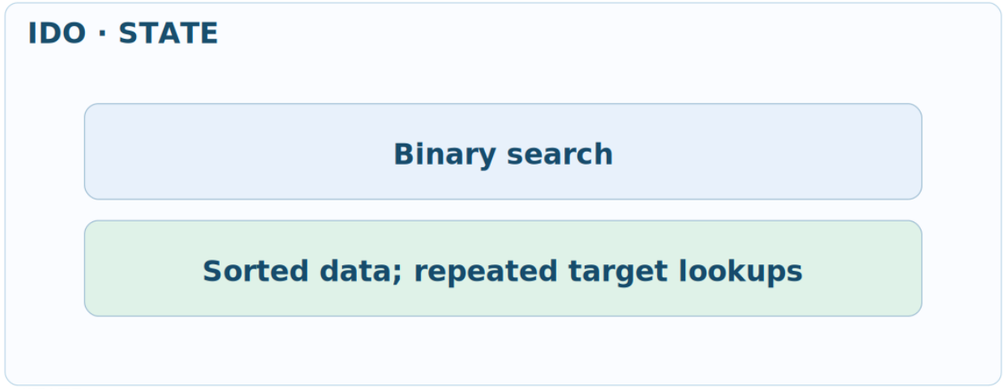
A college searches a million unsorted attendance records once and later queries sorted year-group reports repeatedly. Evaluate algorithms/data processing choices. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose algorithm for 200 sorted pupils by ID with repeated lookups.

Binary search O(log n) is valid because identifiers are sorted; if indexed dictionary is available, keyed lookup is another option.



IDO · STATE

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Choose method for shortest road journey across a network with non-negative costs but no heuristic.
Source code / input:
Dijkstra finds least-cost path with non-negative edge weights; do not confuse with array search.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Identify suitable algorithm for (a) unsorted match (b) sorted match (c) non-negative weighted shortest path. [6]

Question 2

Compare bubble, insertion, merge and quick sort worst-case time and typical auxiliary memory. [8]